

Question	Marking guidance	Mark	Comments
01.1	8;	1	Accept eight
01.2	Phosphodiester (bond);	1	Accept phonetic spellings
01.3	1. DNA helicase – (unwinding DNA and) breaking hydrogen bonds / bonds between chains / bases / strands; 2. DNA polymerase – joins (adjacent) nucleotides OR forms phosphodiester bond / sugar-phosphate backbone;	2	1. Accept H bonds. 1. Accept hydrolyses for breaks 2. Reject forms hydrogen bonds (between nucleotides / bases)
01.4	1. ATP has ribose and DNA nucleotide has deoxyribose; 2. ATP has 3 phosphate (groups) and DNA nucleotide has 1 phosphate (group); 3. ATP- base always <u>adenine</u> and in DNA nucleotide base can be different / varies;	2 max	Both parts of each MP needed 3.Reject Uracil / U 3. Accept C, T or G for different bases Accept annotated diagram for any of the three marks